

HDOC – 8

Question 1:

Write a C program to find the curved surface area and volume of a pipe when inner radius, outer radius, and length are given.

Task1: Understand the Problem Statement:

- As per the given problem statement, we need to create a C program to calculate the surface area and volume of a pipe, while the inner radius, outer radius and the length of the pipe are given
- The input for the program required are:
 - Inner radius in cm
 - Outer radius in cm
 - Length of the pipe in cm
- Output of the program:
 - Curved surface area of the pipe
 - Volume of the pipe
- Formulae for the given problem:
 - Curved surface area of a hollow cylinder = $2 \pi h(R+r)$
Where,
R= outer radius
r= inner radius
h= length of the cylinder
 - Volume of the hollow cylinder = $\pi (R^2+r^2)h$
Where,
R= outer radius
r= inner radius
h= length of the cylinder

Task 2: Standard Test Case Table:

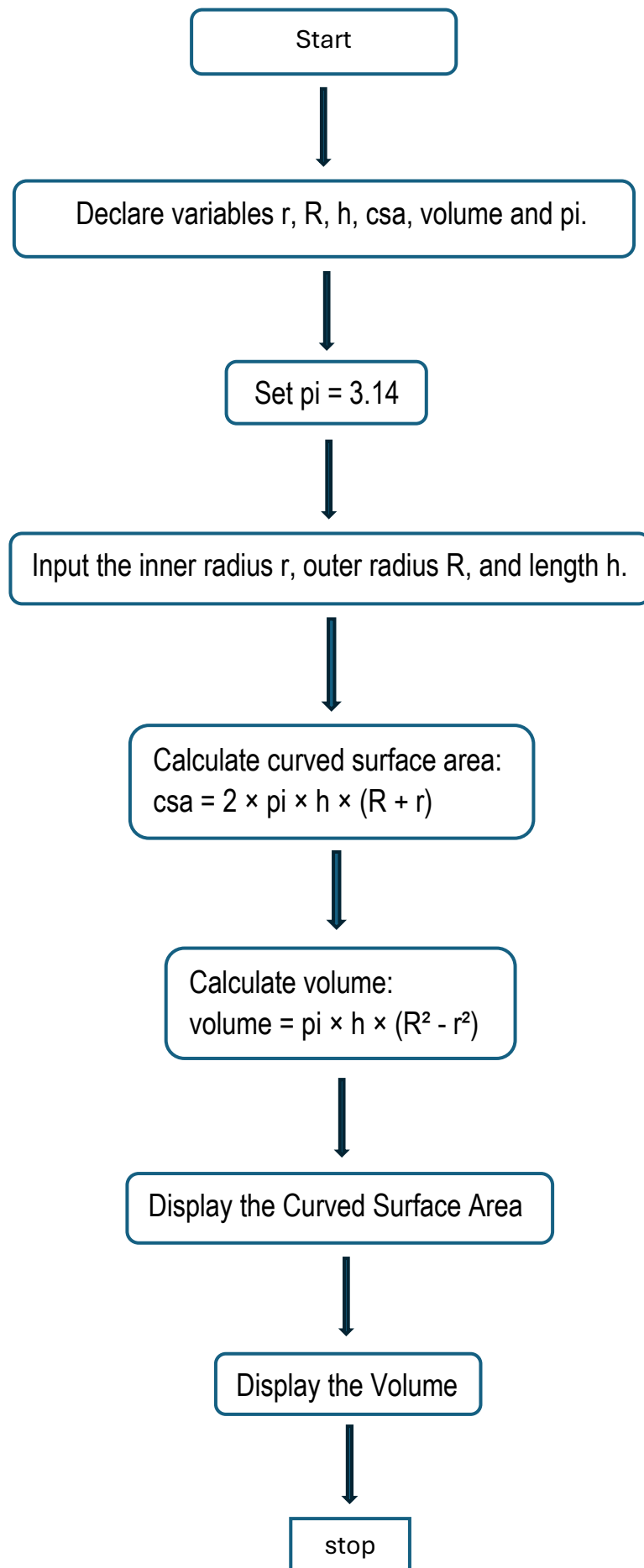
Test Case	Input(s) (Inner Radius, Outer Radius, Length)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	r = 3, R = 7, h = 4	CSA = 251.20 Volume = 502.40	CSA = 251.20 Volume = 502.40	Pass
2	r = 5, R = 10, h = 15	CSA = 1413.00 Volume = 3532.50	CSA = 1413.00 Volume = 3532.50	Pass
3	r = 2.3, R = 4.9, h=10	CSA = 452.16 Volume = 587.81	CSA = 452.16 Volume = 587.81	Pass
4	r = 4, R = 1.5, h = 8	Outer Radius must be greater than inner radius	Outer Radius must be greater than inner radius	Pass
5	r = 3.1, R = 7, h = 9.8	CSA = 621.59 Volume = 1212.11	CSA = 621.59 Volume = 1212.11	Pass

Task 3: The Algorithm:

The algorithm of the C program will be-

- Start
- Declare variables r, R, h, csa, volume and pi.
- Set pi = 3.14
- Input the inner radius r, outer radius R, and length h.
- Calculate curved surface area:
 $csa = 2 \times \pi \times h \times (R + r)$
- Calculate volume:
 $volume = \pi \times h \times (R^2 - r^2)$
- Display the Curved Surface Area.
- Display the Volume.
- Stop

Task 4: The flowchart:



Task 5: C program:

```
#include <stdio.h>

#define PI 3.14

int main()
{
    float innerRadius, outerRadius, length;
    float curvedSurfaceArea, volume;

    // Input
    printf("Enter the inner radius of the pipe: ");
    scanf("%f", &innerRadius);

    printf("Enter the outer radius of the pipe: ");
    scanf("%f", &outerRadius);

    printf("Enter the length of the pipe: ");
    scanf("%f", &length);

}

if (innerRadius >= outerRadius)
{
    printf("\nOuter radius must be greater than inner radius\n");
    return 1;
}

curvedSurfaceArea = 2 * PI * (outerRadius + innerRadius) * length;
volume = PI * (outerRadius * outerRadius - innerRadius * innerRadius) * length;

printf("\n==== Pipe Dimensions =====\n");
printf("Inner Radius      : %.2f units\n", innerRadius);
printf("Outer Radius       : %.2f units\n", outerRadius);
printf("Length              : %.2f units\n", length);
printf("Curved Surface Area : %.2f square units\n", curvedSurfaceArea);
printf("Volume               : %.2f cubic units\n", volume);

return 0;
}
```

Task6: Test Cases Output:

Test case1:

```
Session Actions Edit View Help
(kali@kali)-[~/HDOC8.c]
$ cc ques1.c

(kali@kali)-[~/HDOC8.c]
$ ./a.out
Enter the inner radius(in cm): 3
Enter the outer radius(in cm): 7
Enter the length of the pipe(in cm): 4

=====Pipe dimensions=====
Inner Radius      : 3.00 units
Outer Radius      : 7.00 units
length            : 4.00 units
Curved Surface Area : 251.20 square units
Volume            : 502.40 cubic units
```

Test case2:

```
(kali@kali)-[~/HDOC8.c]
$ ./a.out
Enter the inner radius(in cm): 5
Enter the outer radius(in cm): 10
Enter the length of the pipe(in cm): 15

=====Pipe dimensions=====
Inner Radius      : 5.00 units
Outer Radius      : 10.00 units
length            : 15.00 units
Curved Surface Area : 1413.00 square units
Volume            : 3532.50 cubic units
```

Test case3:

```
(kali@kali)-[~/HDOC8.c]
$ ./a.out
Enter the inner radius(in cm): 2.3
Enter the outer radius(in cm): 4.9
Enter the length of the pipe(in cm): 10

=====Pipe dimensions=====
Inner Radius      : 2.30 units
Outer Radius      : 4.90 units
length            : 10.00 units
Curved Surface Area : 452.16 square units
Volume            : 587.81 cubic units
```

Test case4:

```
(kali@kali)-[~/HDOC8.c]
$ ./a.out
Enter the inner radius(in cm): 4
Enter the outer radius(in cm): 1.5
Enter the length of the pipe(in cm): 8

outerRadius must be greater than innerRadius
```

Test case5:

```
(kali㉿kali)-[~/HDOC8.c]
$ ./a.out
Enter the inner radius(in cm): 3.1
Enter the outer radius(in cm): 7
Enter the length of the pipe(in cm): 9.8

===Pipe dimensions===
Inner Radius      : 3.10 units
Outer Radius      : 7.00 units
length            : 9.80 units
Curved Surface Area : 621.59 square units
Volume            : 1212.11 cubic units
```

Question 2:

Write a C program to find the central angle when the angle at the circumference standing on the same arc is given.

Task1: Understand the Problem Statement:

- The given problem requires a C program to find the central angle when the angle at the circumference standing on the same arc is given.
- The input required:
 - Angle at the circumference (in degrees)
- The output:
 - Central angle (in degrees)
- Formula used:
 - Central Angle = 2 x Angle at Circumference

Task 2: Standard Test Case Table:

Test Case	Input (Angle at Circumference)	Expected Output (Central Angle)	Actual Output	Result
1	30°	60°	60°	Pass
2	45°	90°	90°	Pass
3	60°	120°	120°	Pass
4	90°	180°	180°	Pass
5	15.5°	31.0°	31.0°	Pass

Task 3: The Algorithm:

Step1: Start.

Step2: Declare two floating-point variables:

- circumferenceAngle
- centralAngle

Step3: Read the angle at the circumference from the user.

Step4: If the entered angle is less than or equal to 0:

- Display "Invalid input! Angle must be greater than zero."
- Stop.

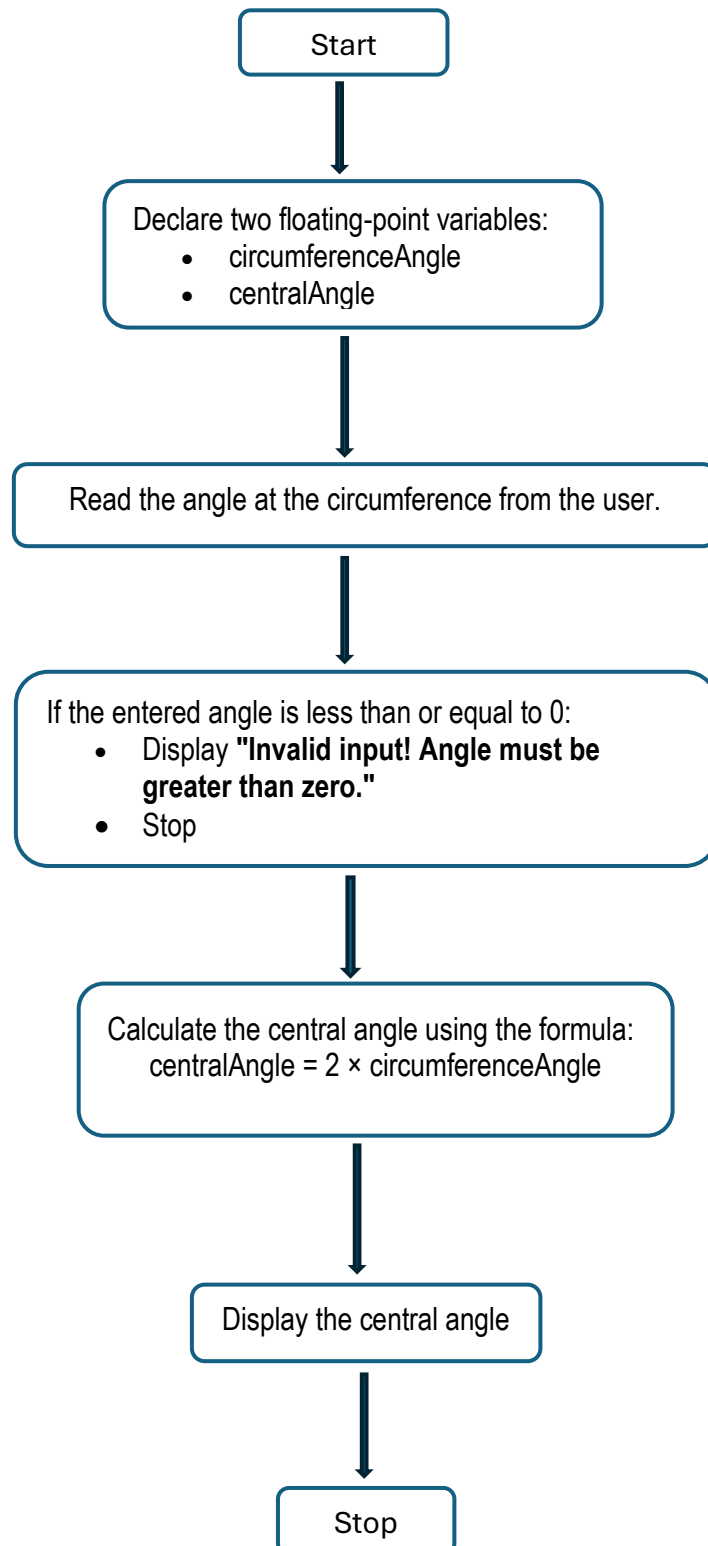
Step5: Calculate the central angle using the formula:

- centralAngle = 2 × circumferenceAngle

Step6: Display the central angle.

Step7: Stop.

Task 4: The flowchart:



Task 5: C program:

```
#include <stdio.h>
int main()
{
    float circumferenceAngle, centralAngle;

    printf("Enter the angle at the circumference (in degrees): ");
    scanf("%f", &circumferenceAngle);

    if (circumferenceAngle <= 0)
    {
        printf("\nError: Angle must be greater than zero.\n");
        return 1;
    }

    centralAngle = 2 * circumferenceAngle;

    printf("\n==== Result =====\n");
    printf("Angle at Circumference    : %.2f degrees\n",
circumferenceAngle);
    printf("Central Angle                : %.2f degrees\n", centralAngle);

    return 0;
}
```

Task6: Test Cases Output:

Test case1:

```
(kali㉿kali)-[~/HDOC8.c]
$ ./a.out
Enter the angle at the circumference(in degrees): 30

=====Result=====
Angle at Circumference      : 30.00 degrees
Central Angle               : 60.00 degrees
```

Test case2:

```
(kali㉿kali)-[~/HDOC8.c]
$ ./a.out
Enter the angle at the circumference(in degrees): 45

=====Result=====
Angle at Circumference      : 45.00 degrees
Central Angle               : 90.00 degrees
```

Test case3:

```
(kali㉿kali)-[~/HDOC8.c]
$ ./a.out
Enter the angle at the circumference(in degrees): 60

=====Result=====
Angle at Circumference      : 60.00 degrees
Central Angle               : 120.00 degrees
```

Test case4:

```
(kali㉿kali)-[~/HDOC8.c]
$ ./a.out
Enter the angle at the circumference(in degrees): 90

=====Result=====
Angle at Circumference      : 90.00 degrees
Central Angle               : 180.00 degrees
```

Test case5:

```
(kali㉿kali)-[~/HDOC8.c]
$ ./a.out
Enter the angle at the circumference(in degrees): 15.5

=====Result=====
Angle at Circumference      : 15.50 degrees
Central Angle               : 31.00 degrees
```